



Software Testing Test Planning

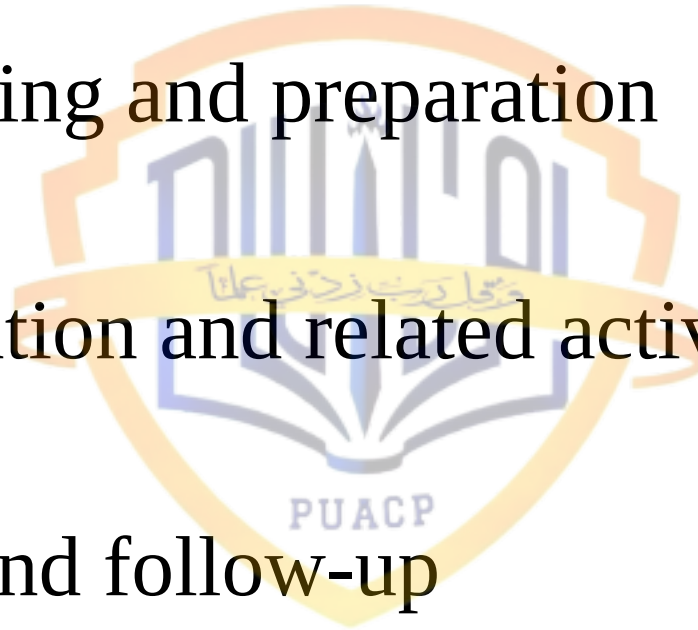
Lecture #30

Recap and Today's Lecture

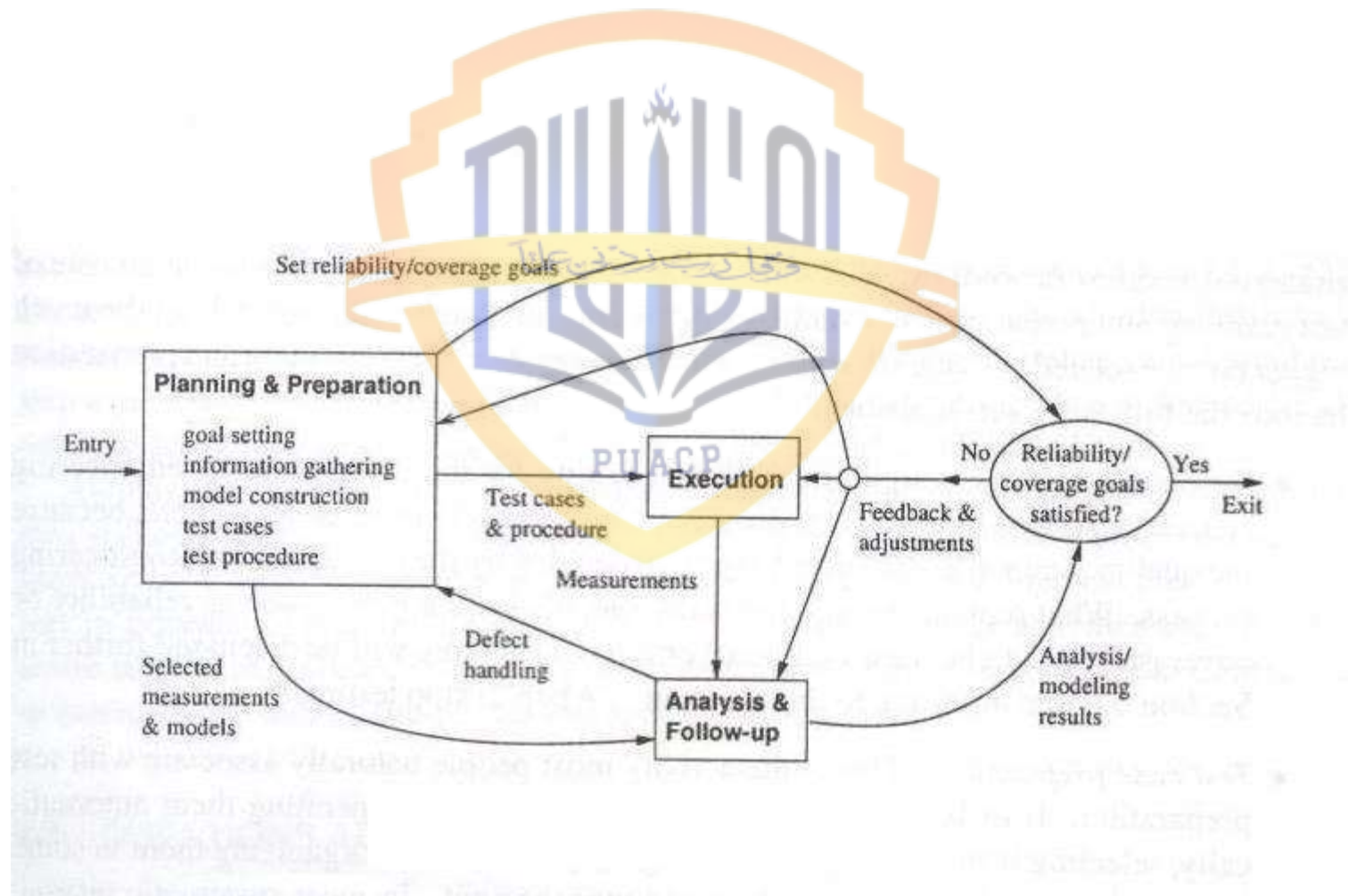
- We have been discussing software testing as a set of activities related to software quality assurance in the last few lectures
- Today, we'll talk about software test planning
- As you know, software systems are becoming complex everyday and we need to plan and prepare ourselves to test them adequately

Generic Testing Process

- Test planning and preparation
- Test execution and related activities
- Analysis and follow-up



Generic Testing Process



Test Planning and Preparation - 1

- As you can see from the picture just shown to you that test planning and preparation is the most important activity in the generic testing process for systematic testing
- Most of the key decisions about testing are made during this stage

Test Planning and Preparation - 2

- Setting of goals for testing
- Selecting an overall testing strategy
- Preparing specific test cases
- Preparing general test procedure

Test Planning - 1

- It is difficult to manage the testing of a large system, considering numerous test cases, problem reporting, and problem fixes
- Just as we plan and manage a project by estimating resources, schedule, cost, and so forth, we must plan the testing of a software product

Test Planning - 2

- The planning activities enable us to do strategic thinking on what types of tests should be executed within the constraints of time and budget
- During the preparation of the test plan, as each test scenario is planned, we must estimate the time it would take to test the system

Test Planning - 3

- The mistake most testers make is **not** planning the test schedule upfront
- They fail to allocate adequate time to test the product and conduct regression testing
- If one of the test dependencies identified upfront in the planning stage is receiving the code on time for testing, the management will realize that if the code does not arrive in testing on time, the entire project schedule will slip

Test Planning - 4

- It's too late to begin test planning when testing actually begins. All that you can do then is hope that baseline designing and unit testing have been so well that things fall into place easily
- This sort of luck is seldom achieved in real projects
- Planning is the only reliable way to mitigate such problems

Test Planning - 5

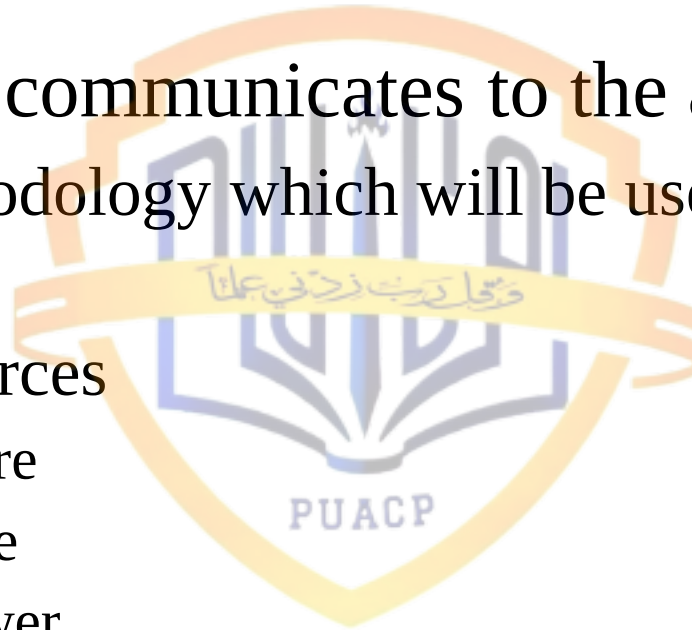
- Beizer claims that test planning can uncover at least as many problems as the actual tests themselves
- This is a very powerful statement based on the observations of a very seasoned and senior researchers in software engineering community

Test Planning - 6

- The test plan acts as a service level agreement between the test department and other significantly impacted departments
- The testing efficiency can be monitored and improved by developing a **through** test plan
- A test plan identifies the **test deliverables**, **functions to be tested**, and **risks**

Test Planning - 7

- A test plan communicates to the audience
 - The methodology which will be used to test the product
 - The resources
 - Hardware
 - Software
 - Manpower
 - The schedule during which tests will be accomplished
 - The process used to manage the testing project

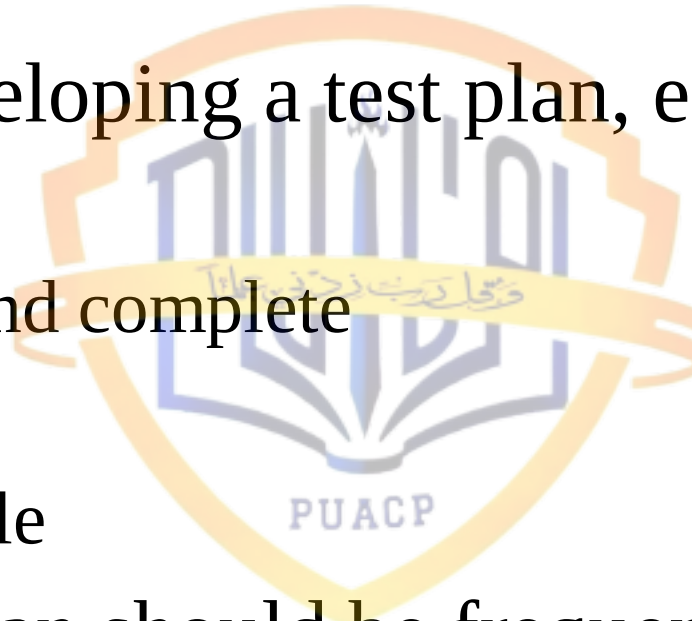


Test Planning - 8

- Before writing a test plan, the following information should be obtained
 - Names of all test analysts
 - Automated test tools to be used for this project
 - Environment required for conducting tests

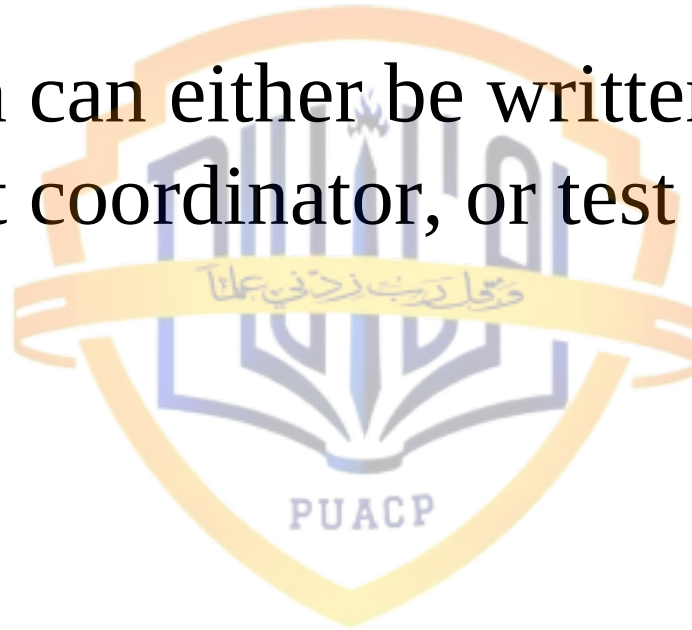
Test Planning - 9

- When developing a test plan, ensure that it is kept
 - Simple and complete
 - Current
 - Accessible
- The test plan should be frequently routed to the appropriate people for feedback and sign-offs



Test Planning - 10

- A test plan can either be written by the test leader, test coordinator, or test manager



Guidelines for Developing Test Plan - 1

- Depending on the features of the system, determine which tests you need to perform
- Once you have identified the types of tests needed to test the system, plan how each of these tests will be performed. Develop test cases
- Use a test plan template (**we'll discuss one today**)

Guidelines for Developing Test Plan - 2

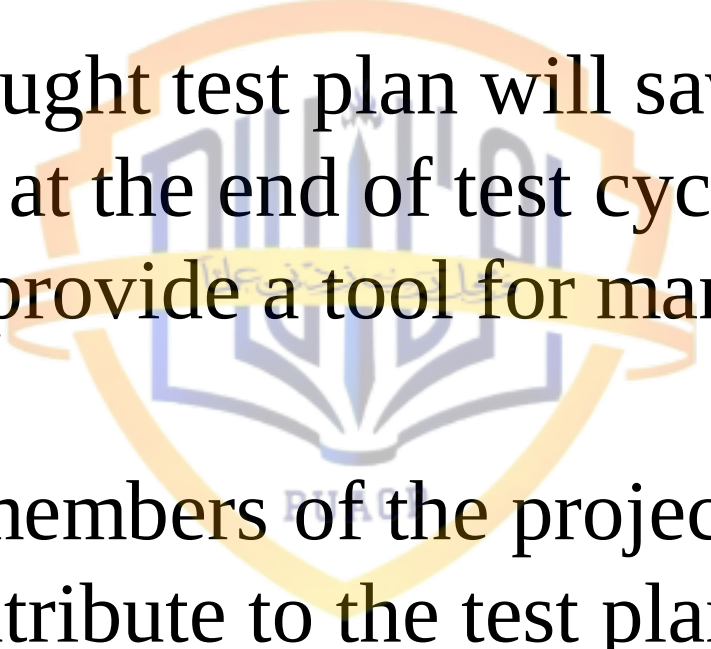
- When the contents of the test plan are finalized, conduct a review for completeness of the test plan
- Incorporate the feedback from the review of the test plan

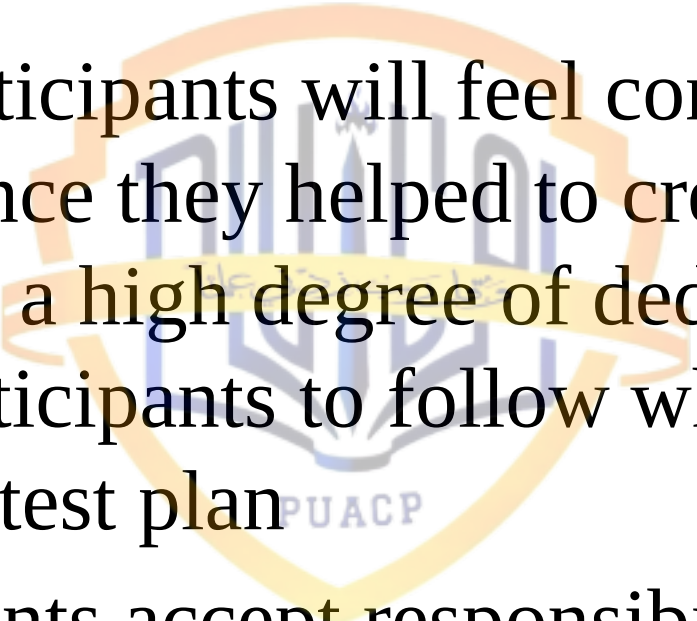
Guidelines for Developing Test Plan - 3

- Get approvals on the contents of the test plan from managers of all the departments that will be impacted by this new product, for example, the managers of **Development, Customer Support, Marketing, Operations, Integrated Technology, Product Assurance, Quality and User Management**

Guidelines for Developing Test Plan - 4

- Follow the test plan to ensure that everyone on the test team uses the process outlined in the test plan and if there are exceptions to this, the deviations are logged as “addenda” to the test plan. For consistency, these addenda should be approved by the same group of individuals who approved the original test plan

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- The logo of the University of Al-Qadisiyah is a circular emblem. It features a stylized open book at the base, with Arabic calligraphy above it. The entire emblem is encircled by a yellow border. The text 'UNIVERSITY OF AL-QADISIYAH' is written in English around the top half of the circle, and 'جامعة القادisiyah' is written in Arabic around the bottom half.
- A well-thought test plan will save hours of frustration at the end of test cycle, reduce risks, and provide a tool for management reporting
 - As many members of the project team should contribute to the test plan document through reviews and discussion as possible

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- All the participants will feel committed to the plan since they helped to create it and this fosters a high degree of dedication among participants to follow what is being said in the test plan
 - If participants accept responsibilities and later do not fulfill them, the test plan provides evidence of the initial agreement

- A test plan provide criteria for measurement, deadlines for test deliverables, and describes procedures for documenting and resolving problems

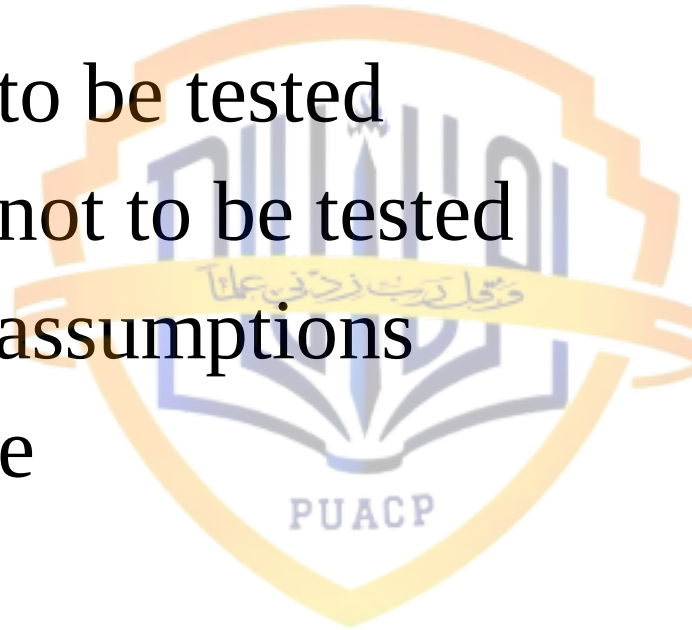
Table of Contents of A Sample Test Plan - 1

- Introduction
- Scope
- Test plan strategy
- Test environment
- Schedule
- Control procedures
- Control activities



Table of Contents of A Sample Test Plan - 2

- Functions to be tested
- Functions not to be tested
- Risks and assumptions
- Deliverable
- Test tools
- Approvals
- Exit criteria



Test Plan: Introduction

- The introduction gives a brief description of the system being tested so anyone reading the test plan would get an overview of the features of the product being tested

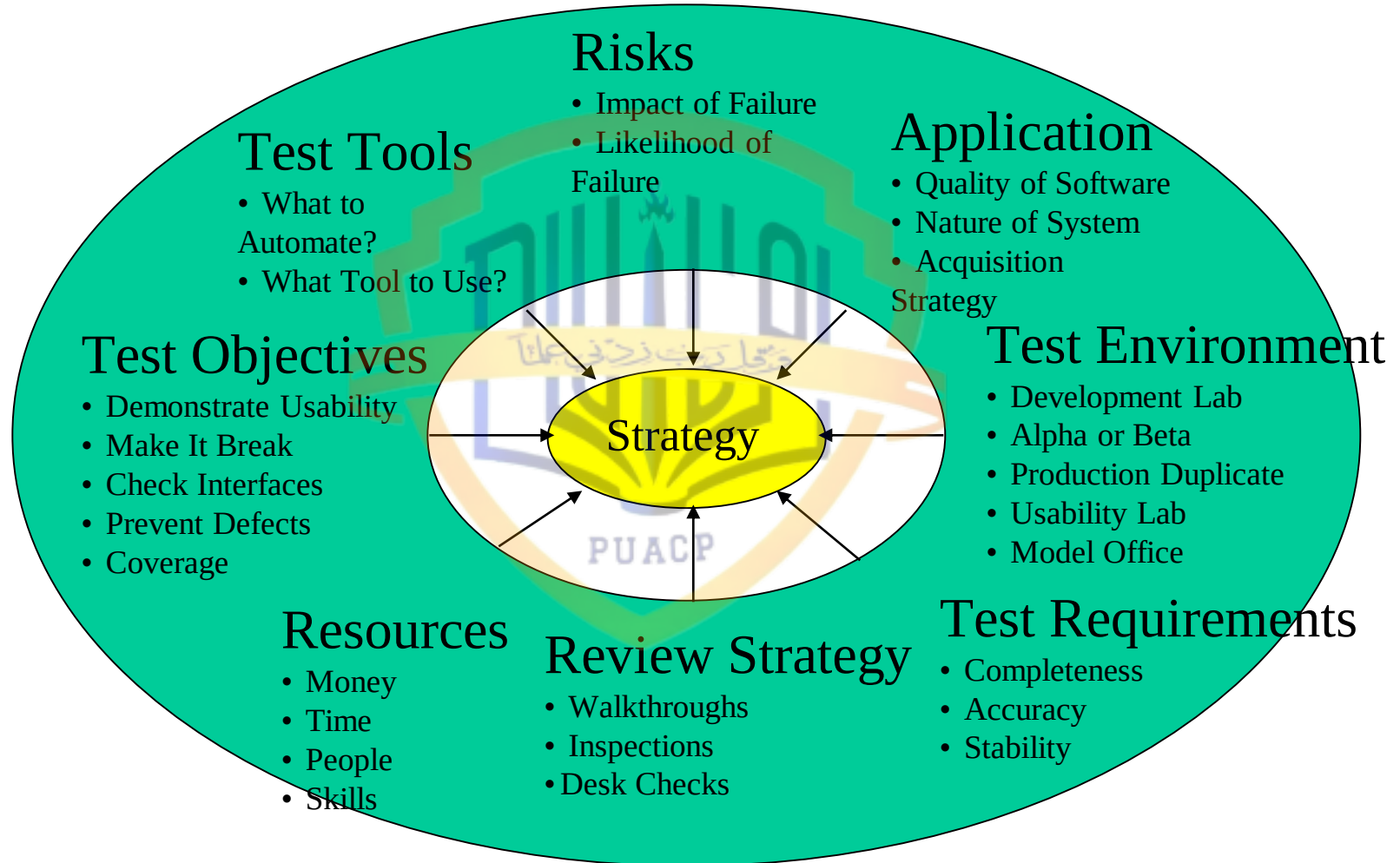
Test Plan: Scope

- The scope describes at a high level what will be tested and what will not be tested in order to avoid ambiguity and identifies what is going to be addressed by the testers

Test Plan: Test Plan Strategy

- Test plan strategy outlines the objectives for developing the test plan and establishes the procedures for conducting the tests
- The strategy should also identify all the tests that will be performed to check the robustness of the product
- A brief description of each test should be given

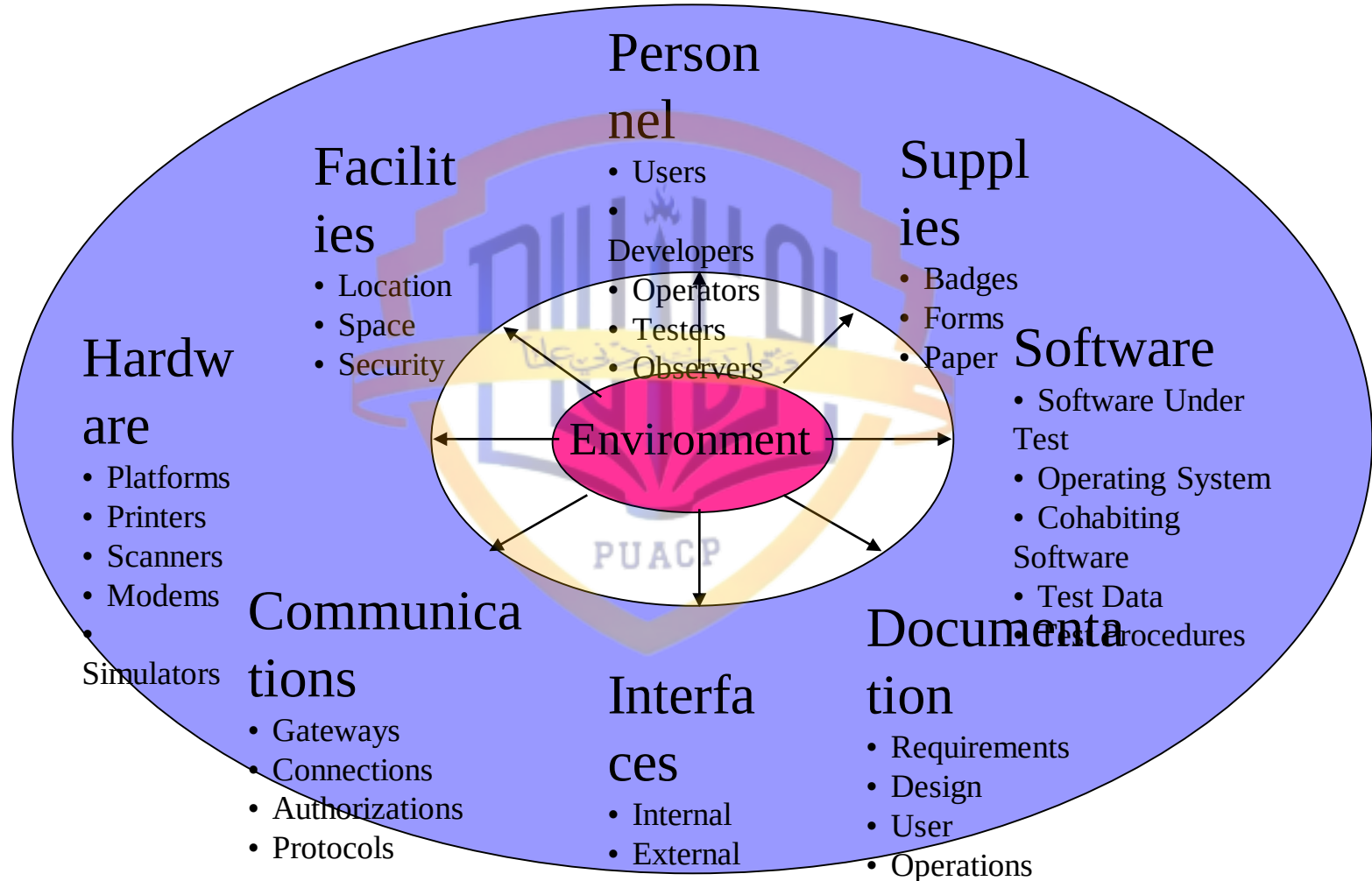
Influences on Test Plan Strategy



Test Plan: Test Environment

- The test environment section lists the hardware and software configurations required to perform tests
- It determines the type of help required from other departments to support the environment tests

Environmental Needs



Test Plan: Schedule

- The overall test schedule is identified in the project plan indicating the start date and end date of test activities. However, detailed schedule of testing should be developed during test planning process, showing start and end dates for
 - Development of the test plan
 - Designing test cases
 - Executing test cases
 - Problem reporting
 - Developing the test summary report

Test Plan: Control Procedures - 1

- This section should have detailed instructions for problem reporting
- A procedure for communicating defects to the development team should be established and published in this section

Test Plan: Control Procedures - 2

- This should include
 - Whether an automated on-line system will be used to record and report problems
 - Whether the problem fixes will be accepted daily or once a week
 - How the problems should be prioritized, for example 1, 2, or 3

Test Plan: Control Activities

- The control activities section indicates when walkthroughs or reviews will be conducted and list the individuals who would participate in these activities

Test Plan: Functions to Be Tested

- This section lists all the functions to be tested
- What you test is more important than how much you test



Test Plan: Functions Not to Be Tested

- This section lists all functions not to be tested, along with the reason for not testing, for example, “low risk” code reused and previously tested with 100% satisfaction

Test Plan: Risks and Assumptions

- This section lists assumptions taken into consideration when developing the test schedule, such as
 - Unit testing will be done prior to receiving code for system testing
 - All testing personnel will have the knowledge of the product and will be experienced testers
 - The computers will be available during all work hours

Test Plan: Deliverable

- The deliverable section lists what will be delivered at the end of test phase, such as,
 - Test plan
 - Test cases
 - Test incident report
 - Test summary report

Test Plan: Test Tools

- This section identifies all test tools which will be used during testing, whether they are in-house or need to be purchased
- Test coverage analyzers might be used to provide information on the thoroughness of one or more test runs of the program
- At the end of the analysis process, the analyzer produces a visual listing of the program with all unexecuted code flagged

Test Plan: Approvals

- This section ensures that the plan is reviewed and approved by the individuals who will either be affected by it, or on whom you are dependent to either receive listing of program with all unexecuted code flagged

Test Plan: Exit Criteria

- This section identifies the exit criteria for the test phase. Some examples of the exit criteria:
 - No priority-one open problems
 - All functions identified in the requirements document are present and working
 - No more than three priority-two problems open

Summary

- Today, we have talked about test planning and discussed the contents of a test plan



References

- Inroads to Software Quality by Alka Jarvis and Vern Crandall (Chapter 6.1-6.3)
- Systematic Software Testing by Rick D. Craig and Stefan P. Jaskiel, Artech House Publishers, 2002 (Chapter 3: Master Test Planning – pictures taken from this chapter)